

Module 2

Management Accounting

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)**Answer 1(a)**

	Physical units	Equivalent units Direct materials	Conversion costs
Opening work in process	8,000		
Started in April	<u>50,000</u>		
Physical units to be accounted for	<u>58,000</u>		
Normal loss	2,000	0	0
Completed units			
- From opening work in progress ((1-80%), (1-60%))	8,000	1,600	3,200
- Started and completed in April (51,000 – 8,000)	43,000	43,000	43,000
Abnormal gain	(1,000)	(1,000)	(1,000)
Closing work in process (70%, 45%)	<u>6,000</u>	<u>4,200</u>	<u>2,700</u>
Physical units accounted for	<u>58,000</u>		
Equivalent units		47,800	47,900
		HK\$	HK\$
Costs incurred in April		119,500	105,380
Costs per equivalent units		2.5	2.2

Answer 1(b)

Process 1 Account					
	Units	HK\$		Units	HK\$
Opening WIP	8,000	28,800	Transferred to Process 2 (W2)	51,000	241,940
Direct materials	50,000	119,500	Normal loss	2,000	0
Conversion costs		105,380	Closing WIP (W3)	6,000	16,440
Abnormal gain (W1)	<u>1,000</u>	<u>4,700</u>			
	<u>59,000</u>	<u>258,380</u>		<u>59,000</u>	<u>258,380</u>

Workings:

(W1) Valuation of abnormal gain
 $1,000 \times \text{HK}\$(2.5 + 2.2) = \text{HK}\$4,700$

(W2) Valuation of completed units to Process 2

	HK\$
Opening WIP	28,800
Costs incurred during April to complete opening WIP	
Direct materials (1,600 × HK\$2.5)	4,000
Conversion costs (3,200 × HK\$2.2)	7,040

Units started and completed in April	
$[(51,000 - 8,000) \times \text{HK}\$(2.5 + 2.2)]$	202,100
	<u>241,940</u>

(W3) Valuation of closing WIP

	HK\$
Direct materials $(4,200 \times \text{HK}\$2.5)$	10,500
Conversion costs $(2,700 \times \text{HK}\$2.2)$	<u>5,940</u>
	<u><u>16,440</u></u>

Answer 1(c)

Normal losses are the losses that are inevitable and expected even in a very efficient process. These could be due to liquids evaporating or wastage during the cutting and forming of raw materials, etc.

When actual losses in a process are greater than the expected normal losses, as a result of inefficient processes, carelessness, lack of training, broken machinery or contaminated raw materials, etc., the difference between the actual losses and the expected normal losses are abnormal losses.

When actual losses in a process are less than the expected normal losses, which could be the result of more efficient new machinery or abnormally effective use of raw materials, etc., the difference between the actual losses and the expected normal losses are abnormal gains.

Answer 2(a)

Relevant costs for the special order:	HK\$
Direct materials – A $(10,000 \times \text{HK}\$12 + 2,000 \times \text{HK}\$10)$	140,000
Direct materials – B $[21,000 \times (\text{HK}\$32,400 \div 1,800 + \text{HK}\$2)]$	420,000
Direct labour $(1,500 \times \text{HK}\$160 \div 2)$	120,000
Variable production overheads $(\text{HK}\$11 \times 20,000)$	220,000
Fixed production overheads	100,000
Selling and distribution overhead	60,000
Income tax $(\text{HK}\$70 \times 20,000 \times 2\%)$	28,000
Opportunity cost for Mazarine $[\text{HK}\$(500 - 320) \div 2 \times 1,500]$	<u>135,000</u>
	<u><u>1,223,000</u></u>

Relevant cost of one unit of Ruby under the special order:
 $\text{HK}\$1,223,000 \div 20,000 = \text{HK}\61.15

As the unit selling price of Ruby under the special order of HK\$70 is higher than its unit relevant cost of HK\$61.15, it is profitable for GCL to take the special order.

Answer 2(b)

Fixed costs are defined as those costs that remain the same, regardless of changes in activity level, over a set time period and production volume range.

Fixed costs can further be categorised into avoidable and unavoidable fixed costs.

Avoidable fixed costs are the fixed costs of the activity of a business that would be avoided if the activity did not exist; thus, it is relevant in short-term decision-making.

On the other hand, unavoidable fixed costs are the fixed costs of the activity of a business that would not be avoided regardless of whether the activity was or was not to exist.

Thus, it is irrelevant in short-term decision-making.

Answer 3(a)

- (i) Direct material used in kg
 $= 30,000 \times 10 \text{ kg} - \text{HK\$}300,000 \div \text{HK\$}15 = 280,000 \text{ kg}$
- (ii) Direct material purchased in kg
 $= (280,000 + 40,000) \text{ kg} = 320,000 \text{ kg}$
- (iii) Direct material purchased in dollars
 $= \text{HK\$}15 \times 320,000 + \text{HK\$}320,000 = \text{HK\$}5,120,000$
- (iv) Actual direct labour cost
 $= \text{HK\$}60 \times 2 \text{ hours} \times 30,000 + \text{HK\$}60,000 - \text{HK\$}122,000 = \text{HK\$}3,538,000$
- (v) Actual direct labour hours used
 $= 2 \text{ hours} \times 30,000 + \text{HK\$}60,000 \div \text{HK\$}60 = 61,000 \text{ hours}$
- (vi) Actual direct labour rate per hour
 $= \text{HK\$}3,538,000 \div 61,000 \text{ hours} = \text{HK\$}58$, or
 $= \text{HK\$}60 - \text{HK\$}122,000 \div 61,000 \text{ hours} = \text{HK\$}58$
- (vii) Actual variable production overheads
 $= \text{HK\$}50 \times 30,000 - \text{HK\$}100,000 = \text{HK\$}1,400,000$
- (viii) Actual machine hours used
 $= 2.5 \text{ hours} \times 30,000 - \text{HK\$}100,000 \div \text{HK\$}20 = 70,000 \text{ hours}$
- (ix) Actual fixed production overheads
 $= \text{HK\$}80 \times 30,000 + \text{HK\$}500,000 = \text{HK\$}2,900,000$

Answer 3(b)

Ideal standards represent optimal levels of performance under perfect operating conditions.

They allow for no machine breakdowns or other work interruptions, and they call for a level of effort that can be attained only by the most skilled and efficient employees working at peak effort continuously.

Currently attainable standards are those standards that are reasonable but challenging.

They allow for normal machine downtime and employee rest periods.

They can be attained through reasonable, but highly efficient, efforts of the average worker.

Currently attainable standards are usually used in setting standards costs as they can signal abnormal conditions and can result in realistic planning and forecasting figures.

Answer 4(a)

	Beauty Division	Spa Division
Return on investment	$\text{HK\$}550,000 / \text{HK\$}5,000,000$ $\times 100\% = 11\%$	$\text{HK\$}1,560,000 / \text{HK\$}12,000,000$ $\times 100\% = 13\%$
Required return (Investment $\times 12\%$)	$\text{HK\$}5,000,000 \times 12\%$ $= \text{HK\$}600,000$	$\text{HK\$}12,000,000 \times 12\%$ $= \text{HK\$}1,440,000$
Residual income ("RI")	$\text{HK\$}550,000 - \text{HK\$}600,000$ $= (\text{HK\$}50,000)$	$\text{HK\$}1,560,000 - \text{HK\$}1,440,000$ $= \text{HK\$}120,000$

Answer 4(b)

Customer perspective

- Customer retention
- Percentage of customers who would recommend the service to a friend

Internal process perspective

- Consumable waste reduction
- Booking accuracy

Learning and growth perspective

- Number of cross-trained employees
- Percentage of employees leaving in less than one year

(other relevant measures are also accepted)

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